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BAP: Bayesian Biomarker-Assisted Platform Design for Dose Ranging in Multi-Agent Multi-Dose Trials

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Abstract

Assessing the long-term benefits of new treatments can be expensive and time-consuming, particularly in disease areas with unmet medical needs. While platform trials enable the evaluation of multiple interventions simultaneously, they currently cannot assess studies involving multiple agents and doses or utilize longitudinal biomarkers in decision-making. We propose a Bayesian biomarker-assisted platform design that offers a unified framework for evaluating multiple investigational agents and their doses in a multi-stage, randomized controlled trial. The design streamlines the drug evaluation process and decreases development costs by including proof-of-concept, futility and superiority monitoring, and dose optimization in a single trial, while avoiding over-allocating patients to a shared placebo or active control arm. To facilitate making real-time interim group sequential decisions, temporarily unobserved long-term responses are estimated from longitudinal biomarker measurements. Design parameters and the maximum sample size are fine-tuned to achieve good frequentist properties. The proposed design is illustrated by a trial of three targeted agents for systemic lupus erythematosus, evaluated by their 24-week response rates. Extensive simulations show that the proposed design compares favorably to several conventional platform designs.

KEY WORDS: Bayesian adaptive design, dose optimization, longitudinal biomarker, platform trial, proof of concept studies.

1 Introduction

Conventional clinical trials investigate new treatments one-at-a-time in a sequential manner. This strategy leads to a particularly long and costly drug development process that is well known to have a low success rate (Dimasi et al., 2013). A more efficient approach is provided by platform designs, which evaluate multiple new agents together with comparison to a control in a single randomized trial with group sequential decisions, rather than in separate trials (Berry, 2015; Woodcock and LaVange, 2017; Saville and Berry, 2016; Bitterman et al., 2020). This setting arises when one or more pharmaceutical companies have developed two or more new agents targeting a particular class of diseases. Platform trials may include statistical rules that select a promising agent, drop agents that are inefficacious or unsafe, or allow new agents to be added to an ongoing study.

Platform trials have been increasingly adopted in real-world clinical research. A recent systematic review by Massonnaud et al. (2025) identified 189 platform trials initiated between January 2015 and August 2024, spanning a wide range of disease areas. Trial initiation peaked during the COVID-19 pandemic but has remained active and increasingly diverse in the post-pandemic period. The review further showed that platform trials are well represented across early-phase trials, including phase I/II or phase II trials (40%), late-phase trials, including phase III trials (41%), and seamless phase II/III trials (19%), although their prevalence varies by disease area. There are many variations of platform trials, depending on factors such as the targeted disease, therapeutic modality, clinical outcomes used to evaluate efficacy and safety, and available resources. Examples of platform designs are given by Wason and Trippa (2014); Wason (2016); Yuan et al. (2016); Ventz et al. (2018); Hobbs et al. (2018); and Polley and Cheung (2019). Most of these designs target phase II settings, where shorter endpoints and smaller sample sizes make adaptive features, including interim arm addition or dropping and adaptive randomization, more feasible to implement.

Existing platform trial designs generally assume that a dose has already been determined for each experimental agent under evaluation. In other words, for an agent to be eligible for inclusion in a platform trial, it is typically expected that a prior phase II dose-ranging study, an essential step in new drug development, has been conducted to identify an acceptably safe and efficacious dose. The literature on dose-ranging designs for evaluating a single agent is extensive. Bornkamp et al. (2007) and Pinheiro et al. (2010) provided overviews on adaptive dose-ranging trial designs, which may estimate a dose–response (DR) function, identify a target dose such as the minimum effective dose (MED), among others. Miller et al. (2007) proposed a Bayesian optimal design based on the widely-used $E_{\max}$ model, using the interim data to adjust the allocation ratio.

Padmanabhan et al. (2010) proposed a design based D -optimality criteria. Ivanova et al. (2012) used a Bayesian isotonic regression approach in a two-stage design to identify an MED. Pinheiro et al. (2014) proposed a model-based dose-ranging design on the basis of multiple comparison procedures and modeling (MCP-Mod) (Bretz et al., 2005). Aouni et al. (2020) provided practical guidelines for designing dose-ranging trials. Mu et al. (2021) proposed a two-stage Bayesian adaptive interval-based design for finding an MED. Other dose-ranging designs are given by Dragalin et al. (2007); Dette et al. (2008); Pinheiro et al. (2010); Miller (2010); Franchetti et al.

(2013); Holland–Letz (2017), among many others. Existing designs have been developed primarily for evaluating a single agent. By the same rationale that motivates platform trials over conventional trials, conducting multiple separate phase II dose-ranging studies prior to entering a platform trial is suboptimal in several respects.

To further reduce trial development costs and expedite the early selection of promising agents for phase III studies, we propose a multi-agent, multi-dose platform design that can simultaneously evaluate multiple doses across several investigational agents in a seamless manner, without the need to conduct separate phase II dose-ranging studies for each eligible agent. In particular, we propose a Bayesian biomarker-assisted platform (BAP) design to screen $L \geq 2$ experimental treatments, each with multiple study doses, compared to a placebo or an active control, C , based on a long-term real-valued response variable, Y , and one or more biomarker processes, Z , observed sooner than Y . Constructing a tailored design for multi-agent multi-dose platform trials in a way that is efficient, scientifically meaningful, and likely to provide patient benefit can be very challenging. The proposed BAP extends existing platform designs in two ways. First, rather than assuming that a dose has been established previously for each experimental agent, the BAP does dose-ranging for each agent. Second, for each patient in the trial who has been treated but not followed long enough to evaluate their response, the BAP uses their earlier observed biomarker process Z to predict their Y . This is done to provide a reliable basis for making interim adaptive decisions that depend on Y during the trial. In fact, long-term evaluation of the primary response endpoint is common in platform trials, with more than 51% of existing trials using evaluation windows longer than 90 days, as reported by Massonnaud et al. (2025).

2 Motivating Trial Example

This work is motivated by AbbVie’s recent phase II trial (NCT03978520), which evaluated three treatment options (upadacitinib, elsubrutinib, and their combination therapy ABBV-599) to improve long-term outcomes in patients with systemic lupus erythematosus (SLE) (Merrill et al., 2024). A total of 341 patients were enrolled and equally randomized to receive either a fixed dose of upadacitinib or elsubrutinib, one of two doses (low vs. high) of ABBV-599, or placebo. The primary endpoint was the change in the SLE Disease Activity Index (SLEDAI), a continuous outcome measured at week 24 post-treatment relative to baseline. The study also collected longitudinal measurements of anti-double-stranded DNA IgG (anti-dsDNA IgG), a biomarker previously shown to be highly correlated with SLEDAI (Han et al., 2022). The trial concluded that both upadacitinib and high-dose ABBV-599 resulted in significant improvements in SLEDAI and substantial reductions in anti-dsDNA IgG levels.

To date, only two drugs, belimumab and anifrolumab-fnia, have received FDA approval for the treatment of SLE. However, long-term outcomes for SLE patients remain unsatisfactory. In response, numerous ongoing trials are evaluating other promising targeted agents, including type I interferon inhibitors, B cell–targeted therapies, and Bruton’s tyrosine kinase inhibitors. As discussed in the Introduction, evaluating multiple agents through separate trials is inefficient and resource-intensive. The NCT03978520 trial should be commended for simultaneously evaluating three experimental agents using a shared placebo in a platform trial format. However, only ABBV-599 was evaluated at multiple dose levels; the doses for upadacitinib and elsubrutinib were predetermined in prior dose-ranging studies. This approach may result in suboptimal

resource utilization and delays in drug development. Furthermore, the NCT03978520 trial adopted a fixed analysis-of-variance (ANOVA) design that treated the five arms as independent and tested the equality of SLEDAI across treatment arms, which was inefficient and lacked adaptability. The proposed BAP design seeks to fill this methodological gap, with the ultimate goal of eliminating the need for separate dose-ranging studies for the experimental agents and enabling more efficient trials through explicit modeling of dose–response relationships.

3 Trial Design Overview

The BAP design can be applied to any randomized phase II multi-agent, multi-dose trial to address the following three key objectives of dose-ranging trials (Bornkamp et al., 2007): (1) determine whether any experimental agent demonstrates a dose–response effect compared to the placebo or active control (C), also known as proof-of-concept (PoC); (2) evaluate whether a clinically relevant improvement (clinical relevance, CR), defined as an increase of at least $\Delta > 0$ in the mean of Y (assuming higher values of Y indicate greater benefit), is likely for at least one dose of agent ℓ within the studied dose range; and (3) identify a target dose, defined as the minimum effective dose (MED) in the SLE trial, although other dose selection criteria may be used depending on the clinical context such as determining the ED90 dose that achieves 90% of the maximum efficacy. The final dose selected for each agent will be evaluated in a subsequent large-scale phase III trial. Therefore, it is especially important to accurately identify a dose that provides a clinically meaningful therapeutic benefit over the placebo or active control, in order to increase the likelihood of success in late-phase trials.

To illustrate the central concept, we present the BAP design in the context of the motivating phase II placebo-controlled SLE trial, in which three experimental agents, each evaluated at multiple doses, are compared with placebo. BAP also applies to trials with active controls without methodological modification. To achieve the three objectives described above, BAP incorporates the following key adaptations within a group-sequential framework:

(1) **Early stopping and selection:** The design allows for early termination of a futile dose or an entire experimental arm if there is insufficient evidence of benefit over the placebo. Conversely, it also enables early selection of a treatment arm and its associated target dose if there is strong evidence of superiority.

(2) **Two-step randomization:** The design employs a two-step randomization scheme to assign newly enrolled patients to either the placebo or an (agent, dose) arm. In the first step, patients are randomized to receive either the placebo or one of the experimental agents using simple randomization. In the second step, those assigned to an experimental arm are adaptively randomized to one of the study doses, with higher probabilities assigned to more promising doses.

The trial objectives and decision framework of the BAP design are illustrated in the flowchart in Figure 1. As the trial progresses, early stopping and selection decisions, as well as within-arm randomization probabilities, are dynamically updated based on the accumulated data. These adaptations generally require timely access to complete outcome data for all treated patients, as a large proportion of missing outcomes can introduce substantial uncertainty or even bias into

decision-making. However, as in the motivating SLE trial, the primary outcome Y often takes a long time to observe (e.g., 24 weeks), and the patient accrual frequently outpaces the outcome observation window. A naïve solution to this issue is to halt patient accrual until sufficient outcome data become available to inform subsequent adaptive decisions. However, this approach would significantly prolong trial duration and poses greater challenges in multi-agent, multi-dose platform trials, where missing data in one experimental arm may delay enrollment or decision-making across the entire platform.

To address this challenge, the BAP design incorporates modeling and decision-making structures that leverage an intermediate biomarker $\mathbf{Z}$ (i.e., anti-dsDNA IgG), which is associated with SLEDAI and measured longitudinally prior to the 24-week response evaluation. The biomarker process is modeled using a Gaussian longitudinal framework, where the mean structure is specified as an additive function of dose and a cubic B-spline function of time, and the residuals follow a first-order autoregressive (AR(1)) correlation structure. The SLEDAI outcome measured at week 24 is also modeled using a Gaussian distribution, with its mean depending on dose and, potentially, selected values from the biomarker trajectory. These submodels are chosen for their flexibility and computational tractability and are tailored to the characteristics of the SLE trial. A comprehensive modeling strategy, along with detailed design considerations, is provided in the following sections. Alternative submodels may be used in other applications, depending on the nature of the outcome and biomarker data.

4 Probability Models

4.1 Multi-Agent Multi-Dose Dose–Response Model

We index the experimental agents by $\ell = 1, \dots, L$, and placebo C by $\ell = 0$. To simplify notation, we assume that all agents have the same number of doses, indexed by $j = 1, \dots, J$. Let $\{d_{\ell,1}, \dots, d_{\ell,J}\}$ denote the raw or standardized doses of each agent ℓ , and denote the dose of C by $d_0 = 0$. Let n be the number of patients treated at some interim point in the trial, and index patients by $i = 1, \dots, n$. Let (a_i, x_i) denote the (agent, dose) combination received by the i^{th} patient, with $x_i \in \{1, \dots, J\}$, $a_i \in \{1, \dots, L\}$, and $x_i = d_0$ if $a_i = 0$.

Our paper’s primary objective is to introduce a new platform design strategy, and therefore, we will not delve into different outcome types or modeling strategies. Instead, we will showcase our design approach within a commonly used setting in the literature. In many existing dose-ranging trials, the primary response variable Y_i is assumed to be a continuous variable following a normal distribution (Pinheiro et al., 2014), that is,

$$Y_i | a_i = \ell, x_i = j \stackrel{\text{iid}}{\sim} N(\mu_{\ell,j}, \sigma_\ell^2), \quad (4.1)$$

where “iid” means *independent and identically distributed*, $\mu_{\ell,j}$ is the mean response for dose j of agent ℓ , and σ_ℓ^2 is the unknown variance of Y_i for agent ℓ . When the outcome is not

normally distributed, one can adopt the two-step approach of Pinheiro et al. (2014), which fits the raw sample estimates of $\mu_{\ell,j}$ using generalized least squares. We numerically evaluate and discuss the sensitivity of our design to data-distribution misspecification in Section 7.4.

While many parametric and non-parametric DR models have been proposed (Bretz et al., 2005; Bornkamp and Ickstadt, 2009; Yuan and Yin, 2011; Gajewski et al., 2019), we adopt the widely-used four-parameter $E_{\max}$ model for $\mu_{\ell,j}$, which is highly flexible and capable of capturing a wide variety of DR curve shapes (Ting, 2006):

$$\mu_{\ell,j} = f(d_{\ell,j}, \theta_{\ell}) = \theta_C + \frac{\theta_{\ell,1} \times d_{\ell,j}^{\theta_{\ell,2}}}{d_{\ell,j}^{\theta_{\ell,2}} + \theta_{\ell,3}^{\theta_{\ell,2}}} \times 1\{\ell \geq 1\}, \quad (4.2)$$

where $1\{A\}$ denotes the indicator function of the event A , $\theta_{\ell} = \{\theta_C, \theta_{\ell,1}, \theta_{\ell,2}, \theta_{\ell,3}\}$, $\theta_C = E(Y | a_i = 0)$ is the baseline effect for C , $\theta_{\ell,1}$ is the maximum effect attributable to drug ℓ , $\theta_{\ell,2}$ is a parameter that controls the steepness of the DR curve, and $\theta_{\ell,3}$ is the dose that produces 50% $\theta_{\ell,1}$. Empirical analyses by Thomas et al. (2014) and Thomas and Roy (2017) have demonstrated that the $E_{\max}$ model can effectively approximate the DR curves for most compounds. It has also been shown to fit SLE trial data well (Furie et al., 2021).

Suppose, temporarily, that the response Y_i can be fully observed for each patient at the time of any interim decision, and let $\mathcal{D}_{\text{com}}(n) = \{a_i, x_i, Y_i\}_{i=1}^n$ denote the resulting complete data based on n treated patients. We take a Bayesian approach to estimate the mean response $\mu_{\ell,j} = f(d_{\ell,j}, \theta_{\ell})$, by assuming the following prior distributions:

$$\begin{aligned} \theta_C &\sim N(\theta_0, \tau_0^2), \quad \sigma_{\ell}^2 \stackrel{\text{iid}}{\sim} \text{IGamma}(\alpha_{\sigma}, \beta_{\sigma}), \quad \ell = 0, \dots, L, \\ \theta_{\ell,1} &\stackrel{\text{iid}}{\sim} N(\theta_1, \tau_1^2), \quad \theta_{\ell,2} \stackrel{\text{iid}}{\sim} \text{TGamma}(\alpha_{\theta}, \beta_{\theta}), \quad \theta_{\ell,3} \stackrel{\text{iid}}{\sim} N_+(\theta_3, \tau_3^2), \quad \ell = 1, \dots, L, \end{aligned}$$

where N_+ is the positively truncated normal distribution, IGamma and TGamma denote the inverse and truncated Gamma distributions, respectively. The hyperparameters in the prior can be determined either based on historical data or specified to ensure a vague prior. We assume a truncated Gamma prior for the parameter $\theta_{\ell,2}$ because an overly large (or small) value would cause numerical issues when computing $d_{\ell,j}^{\theta_{\ell,2}}$. When the complete data $\mathcal{D}_{\text{com}}(n)$ is available and the priors are suitably specified, the posterior distribution $\mu_{\ell,j} | \mathcal{D}_{\text{com}}(n)$ can be sampled using standard Markov chain Monte Carlo (MCMC) methods.

An important advantage of the multi-drug $E_{\max}$ model (4.2) is that the data collected on C can be used to evaluate the L experimental drugs. This approach is more efficient than conducting separate trials for each drug, as it avoids over-allocating patients to placebo C while maintaining

comparable inferential efficiency. In the case that the L experimental drugs have different patient enrollment criteria (e.g., targeting different biomarker subgroups), it may be more appropriate to replace θ_C in model (4.2) with a drug-specific baseline effect $\theta_{\ell,0}$. To leverage information

across biomarker subgroups, a hierarchical structure can be utilized, given by:

$\theta_{\ell,0} \sim N(\theta_C, \tau^2), \ell = 1, \dots, L$, where θ_C is the common mean C effect, and $\tau^2 \sim \text{IGamma}(\alpha_\tau, \beta_\tau)$ is a data-adaptive heterogeneity parameter that regulates the amount of information shared among the L drugs.

4.2 Models for Biomarker and Long-Term Response

In the motivating SLE trial, it takes a substantial amount of time, $T_Y = 24$ weeks, to observe each patient's Y . Consequently, at interim analyses, especially early ones, a large proportion of the n patients enrolled will have follow-up times less than T_Y and thus missing Y values, making the complete data $\mathcal{D}_{\text{com}}(n)$ unavailable. Using only observed outcomes for interim decisions sacrifices information, while waiting for complete follow-up would unacceptably prolong the trial. A major focus of the BAP design is addressing this logistical problem when making interim decisions. Following Jin et al. (2014), Van Lancker et al. (2019), and Van Lancker et al. (2020), we treat the unobserved Y_i values as missing and use informative short-term biomarkers to impute them, thereby enabling efficient and timely interim decisions.

For patient i , let $Z_{i,0}$ denote the baseline biomarker level measured at time $t_{i,0} = 0$ before treatment. Post-treatment biomarker values are measured longitudinally at time points $t_{i,k} \in (0, T_Z]$, with $k = 1, \dots, m_i \leq M$, where m_i is the total number of observations available up to the decision time, M is the maximum number of scheduled post-treatment measurements, and $T_Z \leq T_Y$ denotes the time of the last scheduled biomarker measurement. Define the longitudinal biomarker trajectory for patient i as $\mathbf{Z}_i = (Z_{i,1}, \dots, Z_{i,M})$, where $Z_{i,k}$ denotes the percentage change from baseline at time $t_{i,k}$ for $k \geq 1$.

Under the assumed structural relationships between the (agent, dose) (a_i, x_i) , biomarker $\mathbf{Z}_i$, and long-term response Y_i , illustrated in Figure 1(b), the agent may have a direct effect on Y_i as well as an indirect effect through the drug's effect on $\mathbf{Z}_i$ and the effect of $\mathbf{Z}_i$ on Y_i . Prediction of unobserved Y_i is based on Bayesian parametric models for $[Y_i | x_i, a_i, \mathbf{Z}_i]$ and $[\mathbf{Z}_i | x_i, a_i]$. This allows observed data to be fully used to improve the reliability of interim decisions. While this structure is consistent with a potential mediation pathway, the proposed method does not require a formal causal interpretation or strict mediation assumptions; rather, it relies on the predictive association between $\mathbf{Z}_i$ and Y_i to improve estimation of unobserved outcomes. Therefore, in addition to the main model (4.1)–(4.2), which is constructed for Y_i alone, we assume the following regression model to characterize the structural relationships in Figure 1(b),

$$Y_i | a_i = \ell, x_i = d_{\ell,j}, \mathbf{Z}_i \sim N\left(f(d_{\ell,j}, \theta_\ell) + g(\mathbf{Z}_i, \gamma_\ell), \tilde{\sigma}_\ell^2\right), \quad (4.3)$$

where $f(d, \theta_\ell) = \tilde{\theta}_C + \frac{\tilde{\theta}_{\ell,1} \times d_{\ell,j}^{\tilde{\theta}_{\ell,2}}}{d_{\ell,j}^{\tilde{\theta}_{\ell,2}} + \tilde{\theta}_{\ell,3}} \times 1\{\ell \geq 1\}$ is the $E_{\max}$ model with parameters

$\theta_\ell = \{\tilde{\theta}_C, \tilde{\theta}_{\ell,1}, \tilde{\theta}_{\ell,2}, \tilde{\theta}_{\ell,3}\}$ that are different from θ_ℓ in (4.2), and $g(\mathbf{Z}_i, \gamma_\ell)$ is the effects of biomarker $\mathbf{Z}_i$ on Y_i that may vary across different agents with agent-specific parameters γ_ℓ . If $\mathbf{Z}_i$ has no effect on Y_i , such that $g(\mathbf{Z}_i, \gamma_\ell) = 0$ for all ℓ , the model reduces to the main model (4.1)–(4.2). However, the interpretation of the parameters differs: θ_ℓ represents the conditional effect given $\mathbf{Z}_i$, whereas θ_ℓ in the main model represents the marginal effect. When the biomarker is measured more frequently, model (4.3) can be generalized to a scalar-on-function regression model by incorporating a functional representation of $\mathbf{Z}_i$ (Ramsay and Silverman, 2005; Goldsmith et al., 2011). However, in the motivating SLE trial, the biomarker $\mathbf{Z}_i$ was measured at only a few time points. To ensure robustness and reduce computational complexity, we adopt the simpler model presented here. This choice also aligns with the practical constraints of phase II platform trials, where interim data are typically limited and more flexible models may suffer from overfitting or instability.

In our paper, we take $g(\mathbf{Z}_i, \gamma_\ell) = \gamma_\ell \text{AUC}_i$, where $\text{AUC}_i = \int_0^{t_B} Z_i(t) dt$ denotes the area under the biomarker trajectory curve $Z_i(t)$, quantifying the cumulative biomarker exposure up to time t_B . To model $Z_i(t)$, we propose the following additive longitudinal process:

$$\begin{aligned} Z_{i,k} &= \phi_i(t_{i,k}) + (\lambda_{\ell,1} d_{\ell,j} + \lambda_{\ell,2} d_{\ell,j}^2) 1\{\ell \geq 1\} + \epsilon_{i,k}, \\ \epsilon_{i,k} &= \omega_\ell \epsilon_{i,k-1} + \psi_{i,k}, \quad \psi_{i,k} \sim \text{N}(0, v_\ell^2), \end{aligned} \quad (4.4)$$

where $\epsilon_{i,0} = 0$ and $\omega_\ell \in [-1, 1]$ for $\ell = 0, \dots, L$. In this model, the biomarker measurement $Z_{i,k}$ at time $t_{i,k}$ for patient i is represented as the sum of three components: a patient-specific, time-dependent trajectory function $\phi_i(t_{i,k})$; a fixed drug effect modeled through a quadratic function of dose with parameters $(\lambda_{\ell,1}, \lambda_{\ell,2})$ for $\ell \geq 1$; and a residual term $\epsilon_{i,k}$ that follows a first-order autoregressive (AR(1)) process. This AR(1) structure captures additional time-varying fluctuations not explained by the time trajectory or dose effect, allowing for within-patient correlation in longitudinal biomarker measurements. The condition $\omega_\ell \in [-1, 1]$ is imposed to ensure a stationary AR(1) process. To maintain parsimony and computational feasibility, we do not include interactions between dose and time trajectory functions in the mean structure. Although it would be possible to extend the model to allow for dose–time interactions if the sample size is sufficiently large, such extensions are beyond the scope of this work and are not expected to meaningfully affect the primary decision-making objectives.

We model the patient-specific trajectory function $\phi_{i,\ell}(t_{i,k})$ using a flexible cubic B-spline basis to accommodate nonlinear associations (Brown et al., 2005):

$$\phi_i(t_{i,k}) = \sum_{q=1}^Q \beta_{i,q} b_q(t_{i,k}), \quad k = 2, \dots, M,$$

where $\{b_q(t)\}_{q=1}^Q$ denotes the Q -dimensional set of cubic B-spline basis functions defined on the interval $[0, T_Z]$ with a fixed knot sequence, and $\beta_{i,q}$ represents the patient-specific coefficient corresponding to the basis function $b_q(t_{i,k})$. We assume that the patient-level coefficients follow a random-effects distribution to allow for patient-specific deviations from the average population trajectory; that is, for all patients treated with agent $\ell = 0, \dots, L$, $\beta_{i,q} \sim N(\beta_{\ell,q}, \sigma_{\ell,q}^2)$ ($q = 1, \dots, Q$), where the priors for $\beta_{\ell,q}$ and $\sigma_{\ell,q}^2$ are specified below.

Let $\tilde{\Theta} = \{\theta_\ell, \lambda_{\ell,1}, \lambda_{\ell,2}\}_{\ell=1}^L \cup \{\gamma_{\ell,1}, \gamma_{\ell,2}, \omega_\ell, \beta_{\ell,1}, \dots, \beta_{\ell,Q}, \tilde{\sigma}_\ell^2, v_\ell^2, \sigma_{\ell,1}^2, \dots, \sigma_{\ell,Q}^2\}_{\ell=0}^L$ denote the unknown parameters for the biomarker-assisted model (4.3)–(4.4). We assume the same prior for θ_ℓ and θ_ℓ , and elicit the following prior distributions for the other parameters,

$$\begin{aligned} \lambda_{\ell,1}, \lambda_{\ell,2} &\stackrel{\text{iid}}{\sim} N(\lambda_0, \tau_\lambda^2), \gamma_{\ell,1}, \gamma_{\ell,2} \stackrel{\text{iid}}{\sim} N(\gamma_0, \tau_\gamma^2), \omega_\ell \sim \text{Unif}(-1, 1), \beta_{\ell,1}, \dots, \beta_{\ell,Q} \stackrel{\text{iid}}{\sim} N(\beta_0, \tau_\beta^2), \\ v_\ell^2 &\stackrel{\text{iid}}{\sim} \text{IGamma}(\alpha_v, \beta_v), \tilde{\sigma}_\ell^2 \stackrel{\text{iid}}{\sim} \text{IGamma}(\alpha_{\tilde{\sigma}}, \beta_{\tilde{\sigma}}), \sigma_{\ell,1}^2, \dots, \sigma_{\ell,Q}^2 \stackrel{\text{iid}}{\sim} \text{IGamma}(\alpha_\beta, \beta_\beta). \end{aligned}$$

When no historical information is available for the model parameters, we use non-informative priors by specifying hyperparameters that induce large prior variances.

Let R_i indicate whether the primary outcome Y_i of patient i is observed ($R_i = 1$) or not ($R_i = 0$), and $\mathcal{D}_{\text{obs}}(n) = \{a_i, x_i, R_i, R_i Y_i, Z_{i,1}, \dots, Z_{i,m_i}\}_{i=1}^n$ denote the observed data. To make inferences about the mean response $\mu_{\ell,j}$, we consider two situations: (1) When all the responses are observed with

$\sum_{i=1}^n R_i = n$, such as at the end of the trial, we directly apply the main model (4.1)–(4.2) for

posterior sampling. In this case, no biomarker measurements are used, as the complete primary

outcome data are already observed. (2) If some responses are still pending with $\sum_{i=1}^n R_i < n$ at an

interim analysis, we borrow information from the biomarker-assisted model (4.3)–(4.4) to impute the pending responses, in the spirit of Bayesian data augmentation (Jin et al., 2014), and facilitate timely interim decision-making. A fully Bayesian implementation would repeatedly sample the missing Y_i values from their posterior predictive distributions under the biomarker-assisted model and then update the parameters in the main model using the completed outcome data. To facilitate fast computation in design simulation and calibration, a two-step approximation of this imputation procedure is detailed in the Supplementary Materials. The R code used to implement this two-step approximation via JAGS is also provided in the Supplementary Materials.

Remark 1. Decision-making in the BAP design relies primarily on the main model (4.1)–(4.2). During interim analyses, pending Y_i values are imputed using the biomarker-assisted model (4.3)–(4.4), which leverages partial information from the longitudinal biomarker $\mathbf{Z}$ to support timely decisions without interrupting patient accrual. At the final analysis, because all Y values have been observed, inference is conducted exclusively using the main model and the fully observed outcome data. Thus, the auxiliary model (4.3)–(4.4) is used mainly to guide interim adaptation and does not determine the final dose–response inference. This separation limits the influence of potential auxiliary-model misspecification on the final analysis. Because misspecification of the predictive association between $\mathbf{Z}$ and Y may still affect interim adaptation, its empirical impact is evaluated in the numerical studies in Sections 7.3 and 7.4.

5 Bayesian Predictive Platform Design

5.1 Objective Formulation

Based on the notation introduced in Section 4, recall that μ_C denotes the mean response of the placebo, and $\mu_{\ell,j}$ for $j \geq 1$ represents the effect of the experimental (ℓ, j) combination, where a higher $\mu_{\ell,j}$ indicates greater benefit, as in the SLE trial. We formalize the three trial objectives described in Section 3 within a sequential testing-then-selection framework.

1. *DR Detection:* DR detection for agent ℓ is performed by testing the one-sided hypotheses:

$$H_{\ell,0}^{\text{DR}} : \max_{j \geq 1} \{\mu_{\ell,j}\} \leq \mu_C \quad \text{versus} \quad H_{\ell,1}^{\text{DR}} : \max_{j \geq 1} \{\mu_{\ell,j}\} > \mu_C,$$

where $H_{\ell,0}$ indicates no DR effect and $H_{\ell,1}$ indicates the presence of a DR effect.

2. *CR Establishment:* If $H_{\ell,0}^{\text{DR}}$ is rejected for agent ℓ , the next step is to assess whether a clinically meaningful improvement can be achieved by testing the following hypotheses:

$$H_{\ell,0}^{\text{CR}} : \max_{j \geq 1} \{\mu_{\ell,j}\} < \mu_C + \Delta \quad \text{versus} \quad H_{\ell,1}^{\text{CR}} : \max_{j \geq 1} \{\mu_{\ell,j}\} \geq \mu_C + \Delta,$$

where $\Delta > 0$ denotes a prespecified clinically meaningful improvement over μ_C . Note that establishing CR is more demanding than detecting a DR, since $H_{\ell,1}^{\text{CR}}$ implies $H_{\ell,1}^{\text{DR}}$.

3. *MED Identification:* If $H_{\ell,0}^{\text{CR}}$ is rejected for agent ℓ , the final step is to identify the minimum effective dose (MED) of ℓ , denoted by j_ℓ^{MED} , for use in subsequent phase III trials. The MED is defined as the dose of ℓ for which $\mu_{\ell,j}$ is closest to $\mu_C + \Delta$, formally,

$$j_\ell^{\text{MED}} = \underset{j=1,\dots,J}{\operatorname{argmin}} \left| \mu_{\ell,j} - \mu_0 - \Delta \right|, \text{ with } j_\ell^{\text{MED}} \text{ left undefined if } H_{\ell,0}^{\text{CR}} \text{ is not rejected.}$$

The BAP design evaluates DR and CR using posterior probabilities of the general form:

$$PP_{\ell,j}(\delta, n) = \Pr\{\mu_{\ell,j} \geq \mu_C + \delta \mid \mathcal{D}_{\text{obs}}(n)\}, \quad (5.1)$$

with $\delta = 0$ used for testing $H_{\ell,0}^{\text{DR}}$ versus $H_{\ell,1}^{\text{DR}}$ and $\delta = \Delta > 0$ for testing $H_{\ell,0}^{\text{CR}}$ versus $H_{\ell,1}^{\text{CR}}$.

If both $H_{\ell,0}^{\text{DR}}$ and $H_{\ell,0}^{\text{CR}}$ are rejected based on their corresponding posterior probability rules (described below), the MED is selected as the dose that yields the highest posterior probability of $\mu_{\ell,j}$ falling within a predefined indifference region around $\mu_C + \Delta$. Formally, this posterior probability is defined as

$$PP_{\ell,j}^{\text{MED}}(\Delta, \zeta, n) = \Pr\{\mu_C + \Delta - \zeta \leq \mu_{\ell,j} \leq \mu_C + \Delta + \zeta \mid \mathcal{D}_{\text{obs}}(n)\},$$

where $\zeta > 0$ is a prespecified indifference margin used to define therapeutic equivalence.

5.2 Randomization Scheme

The BAP design makes adaptive decisions in a group-sequential manner across S stages. Let n_s denote the predefined number of patients enrolled by the time of the s th analysis, where $0 < n_1 < \dots < n_{S-1} < n_S = N$, and N is the maximum sample size. To ensure reliable posterior sampling, n_1 should be reasonably large to avoid a high proportion of missing data. For example, n_1 can be chosen based on the accrual rate and the outcome evaluation window so that at least 50% of patients have observed outcomes by the first interim analysis.

There are two randomization steps for each patient. The first randomization step involves selecting from L experimental agents and C using fixed probabilities. A randomization probability ρ is prespecified for C , and the randomization probability for each experimental arm is $(1 - \rho) / L$. The value of ρ can be adjusted to achieve desired patient allocation between C and experimental agents, with a reasonable range from $1 / (JL + 1)$ (i.e., equal allocation among all doses) to $1 / (L + 1)$ (i.e., equal allocation between C and L experimental arms).

In the second randomization step, each patient assigned to agent ℓ in step 1 is further randomized among the J dose levels of that agent. For the first n_1 patients, this within-agent randomization is uniform, with each dose receiving probability $1 / J$. After the first interim analysis, randomization becomes adaptive. To support identification of the MED, we define the adaptive randomization probabilities based on the posterior probability $PP_{\ell,j}^{\text{MED}}$, which is updated sequentially at the end of each stage. We propose allocating more patients to doses that are considered therapeutically equivalent to the MED, with the randomization probability for dose j of agent ℓ defined as:

$$\pi_{\ell,j} = \frac{\{PP_{\ell,j}^{\text{MED}}(\Delta, \zeta, n)\}^\kappa}{\sum_{j'=1}^J \{PP_{\ell,j'}^{\text{MED}}(\Delta, \zeta, n)\}^\kappa}. \quad (5.2)$$

The tuning parameter $\kappa \geq 0$ is calibrated to control the level of sample size imbalance among the doses of agent ℓ , with $\kappa = 0$ corresponding to equal randomization. See, for example, Rosenberger et al. (2001) or Thall and Wathen (2007).

5.3 Group-Sequential Decisions

At each interim analysis, early termination and selection decisions are made based on the updated posterior probabilities $PP_{\ell,j}(\delta, n_s)$ and $PP_{\ell,j}^{\text{MED}}(\Delta, \zeta, n_s)$.

Early Termination for Futility: A combination (ℓ, j) will be considered futile and dropped from the trial at the end of stage s if $PP_{\ell,j}(0, n_s) < c_f(n_s)$, where $c_f(n_s)$ is a prespecified sample size-dependent futility probability cutoff. If all J doses of agent ℓ are dropped, then enrollment to agent ℓ is terminated. This is illustrated by Agent 1 in Figure 1.

Early Selection for Superiority: Let j_ℓ^{CAN} be a *candidate MED* for agent ℓ , defined as

$$j_\ell^{\text{CAN}} = \underset{j=1, \dots, J}{\operatorname{argmax}} \{PP_{\ell,j}^{\text{MED}}(\Delta, \zeta, n_s)\}. \quad (5.3)$$

As illustrated by Agent 3 in Figure 1, the BAP design selects agent ℓ for superiority before reaching the agent-specific maximum sample size of $n_{\max}$ patients, if both of the following conditions are met.

- (1) There exists at least one dose j of agent ℓ such that $PP_{\ell,j}(0, n_s) > c_e(n_s)$, and all remaining dose levels are deemed futile, satisfying $PP_{\ell,j}(0, n_s) < c_f(n_s)$, where $c_e(n_s)$ is a prespecified sample size-dependent superiority threshold.
- (2) The number of patients treated at j_ℓ^{CAN} is greater than a prespecified lower value n^* . This ensures that there will be enough information collected at a final selected MED.

If enrollment to agent ℓ is not halted for futility or superiority after stage $s < S$, BAP continues to randomize patients to agent ℓ in stage $s+1$ using the procedure outlined in Section 5.2. The above decision criteria are adaptively updated at each interim stage for all agents still in the trial until either no experimental agents remain in the trial or the maximum sample size is reached. The entire interim decision-making process of BAP is displayed in Figure S1 of Supplementary Materials.

At either (1) the time of early superiority evaluation or (2) the final analysis, the three trial objectives are addressed using the following rules for each agent $\ell = 1, \dots, L$:

1. *DR Detection*: if $\max_{j \geq 1} \{PP_{\ell,j}(0, n_s)\} > c_e(n_s)$, then $H_{\ell,0}^{\text{DR}}$ is rejected and DR is detected for agent ℓ .
2. *CR Establishment*: if $\max_{j \geq 1} \{PP_{\ell,j}(0, n_s)\} > c_e(n_s)$ and $\max_{j \geq 1} \{PP_{\ell,j}(\Delta, n_s)\} > c_r$, then $H_{\ell,0}^{\text{CR}}$ is rejected and CR is established for agent ℓ , where c_r is a prespecified cutoff for establishing CR. Here, we consider a constant c_r to simplify the design calibration process, but the design can be readily generalized to allow for a sample-size-dependent c_r as well.
3. *MED Identification*: if CR is established for ℓ , then the final estimated MED is the candidate dose j_ℓ^{CAN} ; otherwise, no dose is selected as the MED for agent ℓ .

6 Design Evaluation Criteria

Incorporating dose-ranging into a multi-agent trial and addressing all three trial objectives (i.e., DR detection, CR establishment, and MED identification) simultaneously increases the trial's scope and complexity, making the BAP design more comprehensive than a typical platform trial design. To evaluate the design's performance and calibrate key parameters such as sample sizes and decision cutoffs, we define the core operating characteristics (OCs) aligned with the three trial objectives under a given scenario $\mathcal{H}$, which specifies the true values of all $\mu_{\ell,j}$ and the true associations between Y_i and Z_i . We also outline OC considerations critical to design calibration. Due to the complexity of the BAP design, these OCs must be assessed through simulations across a range of plausible scenarios (FDA, 2020).

6.1 DR Detection Rates

Since DR detection is the first objective to be evaluated, we define the drug-specific DR detection rate for agent ℓ under scenario $\mathcal{H}$ as

$$\Pr(\text{Detect DR in agent } \ell \mid \mathcal{H}) = \Pr(\cup_{s=1}^S \{\max_{j \geq 1} \{PP_{\ell,j}(0, n_s)\} > c_e(n_s)\} \mid \mathcal{H}),$$

which is the probability of rejecting $H_{\ell,0}^{\text{DR}}$ for agent ℓ at any interim or final analysis. Therefore, when $\mathcal{H} = H_{\ell,0}^{\text{DR}}$, this corresponds to the drug-specific type I error rate for DR detection, and when $\mathcal{H} = H_{\ell,1}^{\text{DR}}$, it corresponds to the *drug-specific power for DR detection* ($\text{DSP}_\ell^{\text{DR}}$). A desired BAP design should satisfy the following two criteria for DR detection:

Criterion (1): $\text{FWER}^{\text{DR}} \leq \alpha^{\text{DR}}$, where FWER^{DR} is the family-wise type I error for DR detection under $\cap_{\ell=1}^L H_{\ell,0}^{\text{DR}}$, and α^{DR} is a prespecified nominal value.

Criterion (2): $\text{DSP}_{\ell}^{\text{DR}} \geq 1 - \beta^{\text{DR}}$, where β^{DR} is a prespecified nominal value.

The DR detection rates are primarily determined by the sample size N and probability cutoffs $c_e(n_s)$ and $c_f(n_s)$. Inspired by Shi and Yin (2019) and mimicking the O'Brien–Fleming α -spending function (O'Brien and Fleming, 1979; DeMets and Lan, 1994), we specify the functional form for $c_e(n_s)$ as $c_e(n_s) = 2\Phi(z_{\alpha'} / \sqrt{n_s / n_s}) - 1$, where $\Phi(\cdot)$ is the standard normal, $N(0,1)$, cumulative distribution function and $z_{\alpha'}$ is the α' th percentile of $N(0,1)$. In contrast to Shi and Yin (2019), where α' was set to $\alpha^{\text{DR}} / 2$, BAP requires α' to be determined numerically due to multiplicity, correlations between the control and (agent, dose) effects, and adaptive randomization. To reduce the burden of parameter calibration, we simply take $c_f(n_s) = 1 - c_e(n_s)$, so that BAP is equally likely to claim futility or efficacy under $H_{\ell,0}^{\text{DR}}$.

The comprehensiveness of using both long-term endpoints and longitudinal biomarkers in the BAP leads to numerous potential null and alternative scenarios. It is thus technically infeasible to precisely control FWER^{DR} and $\text{DSP}_{\ell}^{\text{DR}}$. To address this, we propose to numerically search for the values of α' and N under special $H_{\ell,0}^{\text{DR}}$ and $H_{\ell,1}^{\text{DR}}$, where the joint model for Y and $\mathbf{Z}$ is explicitly specified. In particular, under the biomarker-assisted model (4.3)–(4.4), we set $\tilde{\theta}_{\ell,1} = 0$ and $g(\mathbf{Z}_i, \gamma_{\ell}) = 0$ (or equivalently, $\gamma_{\ell} = 0$ when $g(\mathbf{Z}_i, \gamma_{\ell}) = \gamma_{\ell} \text{AUC}_i$). We refer to this special null hypothesis $H_{\ell,0}^{\text{DR}}$ as the *independent null* for DR detection, under which the unconditional expectation of Y_i is flat for any ℓ , and Y_i is uncorrelated with $\mathbf{Z}_i$.

Remark 2. Heuristically, the independent null yields a less concentrated posterior for $\mu_{\ell,j} - \mu_C \mid \mathcal{D}_{\text{obs}}(n)$ compared to null scenarios where Y_i is correlated with $\mathbf{Z}_i$, as it asymptotically borrows no information from $\mathbf{Z}_i$ to impute the missing Y_i during the imputation step. As a result, calibrating α' under the independent null may yield conservative control of FWER^{DR} for some null scenarios in which Y_i is correlated with $\mathbf{Z}_i$. This pattern is supported by the sensitivity analyses presented below.

Similarly, we follow the theory of ranking and selection (Thall et al., 1988) and specify the *least favorable alternative* (LFA) $H_{\ell,1}^{\text{DR}}$, where $\mu_{\ell,1} = \dots = \mu_{\ell,J-1} = \mu_0$ and $\mu_{\ell,J} = \mu_0 + \Delta$. In other words, we assume a step-wise DR curve under $H_{\ell,1}^{\text{DR}}$, with the highest dose J being the MED, which also violates the assumed E_{max} functional form. In addition, consistent with the rationale in Remark 2, we assume that Y_i is correlated with $\mathbf{Z}_i$ under this LFA. As a result, the LFA is

constructed to minimize $\text{DSP}_\ell^{\text{DR}}$ among alternative scenarios where either the monotonicity condition holds or Y_i is correlated with $\mathbf{Z}_i$.

6.2 CR Establishment and MED Identification Accuracy

Since MED identification is contingent upon the establishment of clinical relevance (i.e., an MED can only be identified when both DR and CR are established), we focus on a composite index, the drug-specific MED identification accuracy, defined as the joint probability of

$$\Pr(\text{Reject } H_{\ell,0}^{\text{DR}} \text{ and } H_{\ell,0}^{\text{CR}}, \text{ and } j_\ell^{\text{CAN}} = j_\ell^{\text{MED}} \mid \mathcal{H}),$$

where j_ℓ^{CAN} denotes the dose selected by the design, and j_ℓ^{MED} is the true minimum effective dose for agent ℓ . Under a scenario where agent ℓ does not satisfy the CR criterion (denoted as $H_{\ell,0}^{\text{CR}}$), this index corresponds to the *drug-specific type I error rate for establishing CR* (DSER $_{\ell}^{\text{CR}}$). Conversely, when a clinically meaningful MED exists for agent ℓ (denoted as $H_{\ell,1}^{\text{CR}}$), the MED identification accuracy can be interpreted as the *drug-specific generalized power* (DSGP $_{\ell}^{\text{CR}}$), which reflects the probability of making all correct decisions across the three main objectives. A desired BAP design should satisfy the following two criteria for CR establishment and MED identification:

Criterion (3): $\text{DSER}_{\ell}^{\text{CR}} \leq \alpha^{\text{CR}}$, where α^{CR} is a prespecified nominal value.

Criterion (4): $\text{DSGP}_{\ell}^{\text{CR}} \geq 1 - \beta^{\text{CR}}$, where β^{CR} is a prespecified nominal value.

Given the probability cutoffs $c_e(n_s)$ and $c_f(n_s)$ are determined to control the DR detection rates, the MED identification accuracy is mainly determined by the remaining design parameters N , n^* , and c_r . In general, a larger n^* provides more information at the estimated MED, thus a higher DSGP $_{\ell}^{\text{CR}}$. Because we set c_r to be a constant, its value can be calculated easily based on simulated samples of $\text{PP}_{\ell,j}(\Delta, n_s)$ after determining the values of $\{\alpha', N, n^*\}$. We use the LFA to construct $H_{\ell,1}^{\text{CR}}$. To construct $H_{\ell,0}^{\text{CR}}$, a practical formulation of is given by

$$H_{\ell,0}^{\text{CR}} : \mu_{\ell,1} = \dots = \mu_{\ell,J} = \mu_0 + \Delta/2, \quad (6.1)$$

such that the highest dose falls short of the clinically meaningful effect size by $\Delta/2$. The above CR null (6.8) can be adjusted in practice to meet the clinical requirements specified by the investigators. Similar to the previous section, we assume that Y_i and $\mathbf{Z}_i$ are uncorrelated under both $H_{\ell,0}^{\text{CR}}$ and $H_{\ell,1}^{\text{CR}}$ in order to construct the least concentrated posterior distribution of $\mu_{\ell,j} - \mu_C \mid \mathcal{D}_{\text{obs}}$, thereby ensuring a conservative design calibration.

7 Application to the SLE Trial

7.1 Trial Setting

By refitting the observed data from the NCT03978520 trial (Merrill et al., 2024) using parametric functions, we illustrate how to implement the BAP design in a placebo-controlled platform SLE trial that evaluates three agents, each with four standardized doses (0.1, 0.3, 0.6, 1.0). This results in a total of $12+1=13$ unique (agent, dose) combinations. All the three experimental agents are available at the start of the trial, and no new agents will be introduced thereafter. The primary endpoint Y is defined as the change in SLEDAI at week 24 post-treatment. Based on Merrill et al. (2024), $\mu_0 = E(Y | a_i = 0)$ under the placebo arm is approximately 3 with $\sigma^2 = \text{Var}(Y_i | a_i = 0) = 9$. The clinically meaningful difference is set at $\Delta = 2$, indicating that any experimental agent that can yield a change in SLEDAI of $3+2=5$ at week 24 is considered promising. The biomarker $\mathbf{Z}_i$ represents the percent change in anti-dsDNA IgG level from baseline, measured at weeks $t = 1, 2$, and then every two weeks through week 16 (i.e., $t = 4, 6, \dots, 16$), yielding a total of nine longitudinal measurements per patient. The actual measurement time $t_{i,k}$ (in weeks) for the k th measurement of patient i is assumed to be uniformly distributed within ± 3 days of the scheduled time. Based on data from the NCT03978520 trial, the baseline anti-dsDNA IgG level is assumed to follow a uniform distribution, $Z_{i,0} \sim \text{Unif}(15, 25)$. The underlying time-dependent trajectory function $\phi_\ell(t_{i,k})$ (i.e., the percent change) is specified as

$$\phi_\ell(t_{i,k}) = 1 - \exp(-t_{i,k} / 4), \quad 1 \leq t_{i,k} \leq 16.$$

We assume a unit variance for $Z_{i,k}$ and vary the conditional variance of $Y_i | x_i, a_i, \mathbf{Z}_i$ in (4.3) such that the unconditional variance of Y_i is 9. The biomarker trajectory function and DR curves under the null and alternative scenarios used to calibrate BAP are shown in Figure S2 of the Supplementary Materials. As explained in Section 6, to calibrate the BAP design, the null and alternative scenarios assume no correlation between Y_i and $\mathbf{Z}_i$. In a later section, we assess the robustness of the design by generating biomarker trajectories as well as assuming various association structures between Y_i and $\mathbf{Z}_i$ that violate the model assumption in (4.4).

To implement the BAP design, the investigators have specified the following trial parameters: the nominal levels of FWER^{DR} and $\text{DSER}_\ell^{\text{CR}}$ are $\alpha^{\text{DR}} = .05$ and $\alpha^{\text{CR}} = .10$. The nominal level for $\text{DSP}_\ell^{\text{DR}}$ should be controlled at $1 - \beta^{\text{DR}} \geq 0.7$, and the nominal level for $\text{DSGP}_\ell^{\text{CR}}$ should be controlled at $1 - \beta^{\text{CR}} \geq 0.5$ under the LFA. The fixed allocation ratio between placebo and agents is $\rho = .20$, and the tuning parameter for the within-agent AR probability is $\kappa = 1$. The investigators have also decided to conduct three interim looks when 25%, 50%, and 75% of the total patients have been treated in the trial, followed by a final analysis once the 24-week response is available for all treated patients. Patient accrual is expected to follow a Poisson process, with approximately 40 patients enrolled over 6 months.

7.2 Design Calibration

Based on the configuration of the cubic spline and prior specification detailed in the Supplementary Materials, we use a two-phase search algorithm to determine the design parameters $\{\alpha', c_r, N, n^*\}$ of the BAP design, where the agent-specific maximum sample size $n_{\max}$ is fixed at $n_{\max} = (1 - \rho)N / L$. Preliminary simulations show that BAP's operating characteristics are very close to those of its complete-data counterpart, BAP_C , where decisions are based on complete response data without additional biomarker-assisted modeling, so fewer parameters are involved. To reduce computational complexity, we identify a suitable set of parameters for BAP_C in the first phase, and then fine-tune the parameters for BAP in the second phase. Using the Seadragon high-performance computing cluster, we ran 1,000 parallel jobs, each simulating 10 trials. The first phase, which involved relatively simple posterior sampling, was completed within an hour. The second phase, focused on fine-tuning, took approximately six hours, with 10,000 simulations completed in under an hour.

Step 1. The FWER^{DR} for DR detection generally is not sensitive to the values of N and n^* . In a preliminary search, one can identify a suitable value of α' based on an initial guess of (N, n^*) under the constraint $\text{FWER}^{\text{DR}} \approx \alpha^{\text{DR}}$. In the SLE trial, we initially set $(N, n^*) = (300, 30)$ and identified $\alpha' = 0.10$ for BAP_C . Based on 10,000 simulated trials, the resulting FWER^{DR} for BAP_C with $\alpha' = 0.10$ was 0.051 under the global null.

Step 2. To achieve desired $\text{DSP}_\ell^{\text{DR}} \geq 1 - \beta^{\text{DR}}$ and accurate MED estimation, we considered different values of (N, n^*) : $N \in \{300, 330, \dots, 480\}$ and $n^* \in \{20, 25, 30, 35, 40\}$, leading to $7 \times 5 = 35$ possible combinations. With $\alpha' = .10$, two sets of (N, n^*) , $(360, 30)$ and $(420, 30)$, were identified, leading to $\text{DSP}_\ell^{\text{DR}} = .75$ and $.80$, respectively. The dose estimation accuracy $\Pr(j_\ell^{\text{CAN}} = j_\ell^{\text{MED}})$ was at least .95 for both sets. After consulting with the investigators, we finally chose $(N, n^*) = (360, 30)$ for BAP_C with $n_{\max} = 96$.

Step 3. Taking $(\alpha', N, n_{\max}, n^*) = (.10, 360, 96, 30)$ identified for BAP_C in Steps 1 and 2 as initial values for the BAP design, we performed a fine-tuning of α' to ensure that BAP maintains its FWER^{DR} at the nominal level α^{DR} . This fine-tuning process required the generation of longitudinal biomarkers and missingness indicators in a simulation, which is detailed in the Supplementary Materials. For the SLE trial, $\alpha' = .10$ was selected, leading to $\text{FWER}^{\text{DR}} = .05$. Based on $(\alpha', N, n^*) = (.10, 360, 30)$, the drug-specific power was $\text{DSP}_\ell^{\text{DR}} = .72$, which is within the prespecified range of .70 to .80. Finally, we selected the value $c_r = .19$ so that the type I error for CR establishment $\text{DSER}_\ell^{\text{CR}}$ was controlled at the level of .10 under the null (6.8) and the dose-specific generalized power $\text{DSGP}_\ell^{\text{CR}}$ is 0.5 under the LFA.

7.3 Numerical Studies

Simulation studies were conducted to evaluate the performance of BAP under various scenarios, including comparisons with other designs such as the one-stage analysis of variance design (ANOVA), the multi-stage Isotonic design (Iso), the naive version of BAP (Naive), an equal-randomization version of BAP (ER), and BAP_c . Details of these comparator designs are provided in the Supplementary Materials. We considered a total of twelve scenarios encompassing various DR curve shapes, different biomarker–response relationships, and multiple forms of model misspecification. Detailed data-generating procedures are provided in Table S1, and the corresponding DR curves are illustrated in Figure S3 of the Supplementary Materials. It is important to note that scenarios 7–12 involve various situations of model misspecification, including (1) the main model for Y not being an $E_{\max}$ function, (2) the trajectory function for Z being incorrect, and (3) the dependence between Y and Z being non-linear and non-additive, among others. The numerical results for the core OCs described in Section 6 are computed based on 10,000 simulated trials for each design under each scenario.

- *DR Detection Rates:* Tables 1 and 2 present the DR detection rate, average sample size, and average trial duration. Scenario 1 is the special null scenario used in design calibration. Due to calibration, all designs are expected to control $FWER^{DR}$ in this scenario. Scenarios 2 and 3 consider other null scenarios where Y_i is correlated with Z_i via AUC_i : in scenario 2, the correlation between Y_i and AUC_i is 0.5, whereas in scenario 3, that is 1. ANOVA, Iso, and Oracle designs only use complete Y_i , therefore, their performance is not affected by different associations between Y_i and Z_i . Although the proposed design relies on Z_i , as shown in Table 1, $FWER^{DR}$ is well controlled in these null scenarios. In scenarios 4–12, a DR effect is present for some or all agents. As shown in Tables 1 and 2, the BAP design performs comparably to the oracle BAP_c and outperforms other designs in terms of drug-specific DSP_ℓ^{DR} , provided that a DR effect exists for the corresponding agent.

- *CR Establishment and MED Identification Accuracy* The MED identification accuracy (or generalized power) of the six methods is given in Tables S2 and S3 in the Supplementary Materials. We omit the results for scenarios 1 – 3, which represent global null settings for DR detection, as all methods achieve over 99% accuracy in correctly identifying the absence of CR. Scenarios 4 and 5 are null scenarios for CR establishment, with scenario 4 used in design calibration and scenario 5 incorporating a correlation between Y_i and Z_i . All designs robustly control $DSER_\ell^{CR}$ regardless of the presence or absence of this correlation. In scenarios 6 – 12, where CR and thus the MED exist for some agents, the BAP design demonstrates superior performance, i.e., achieving a higher average rate of successful CR establishment (i.e., one minus the NA rate) and more accurate identification of the correct MED, compared to the ANOVA, Iso, Naive, and ER designs. Similarly, when CR is truly absent for an agent, the BAP design maintains a high probability of correctly identifying the absence of CR.

Because BAP incorporates adaptive randomization and real-time interim go/no-go decisions, it is also important to evaluate its efficiency in terms of patient allocation and overall trial duration. These metrics are described and compared in the Supplementary Materials.

7.4 Robustness under Model Misspecification

The BAP design relies on several modeling assumptions whose violation may affect performance in different ways. The main model (4.1)–(4.2) assumes a Gaussian distribution for the primary outcome Y and an $E_{\max}$ function for the dose–response relationship. The biomarker-assisted model (4.3)–(4.4) further assumes a quadratic dose–biomarker relationship, a Gaussian longitudinal structure for $\mathbf{Z}$, and an additive $\mathbf{Z}$ – Y association mediated through the AUC of $\mathbf{Z}$. As shown in Table S1, many data-generating mechanisms in Section 7.3 already violate one or more of these assumptions, including step-wise or non-smooth dose–response curves and non-quadratic dose–biomarker relationships. To further assess sensitivity to model misspecification, we performed additional numerical studies using the nine data-generating mechanisms given in Table S6, applied under scenario 2, a non-independent null $H_{\ell,0}^{\text{DR}}$, and scenarios 8 and 12. These mechanisms include non-Gaussian distributions for both Y and $\mathbf{Z}$ and varying associations between Y and $\mathbf{Z}$, induced through different $\mathbf{Z}$ – Y link functions or different variances of $\mathbf{Z}$, while holding the marginal mean and variance of Y_i fixed for each agent–dose combination within each scenario.

As shown in the Supplementary Materials, BAP has stable operating characteristics under the non-Gaussian distributions considered and maintains good control of FWER^{DR} across the data-generating mechanisms evaluated. This empirical stability is partly attributable to the numerical calibration of the design parameters under the independent null and to the fact that, at the moderate sample sizes considered, posterior inference for $\mu_{\ell,j}$ is driven primarily by the marginal mean and variance of Y . Therefore, in the scenarios considered, the design performance appears relatively insensitive to moderate deviations from the Gaussian assumption. In contrast, the efficiency gain from biomarker borrowing, such as improved DR detection power or reduced average sample size, depends on the extent to which the auxiliary model captures useful predictive information about Y . Misspecification of the auxiliary model can therefore affect interim adaptation through less accurate imputation of pending outcomes, even though final dose–response inference remains based on fully observed primary outcomes.

7.5 Alternative Trial Configurations

In the Supplementary Materials, we evaluated the performance of the BAP design under alternative trial configurations without recalibrating the design parameters. In each configuration, a single trial parameter was modified from its originally specified value, while all other trial settings were held constant: (a) five interim analyses with 90 patients enrolled in the first stage and 54 in each subsequent stage; (b) indifference margin $\zeta = 0.50$; (c) accrual rate of 80 patients per 6 months; (d) fixed allocation ratio $\rho = 1/7$; and (e) first-stage sample size $n_1 = 135$ with 75 patients in each of the following three stages. The results suggest that BAP maintains good type I error control and achieves competitive efficiency under the alternative configurations considered, even though these configurations differ from those used for calibration. Additionally, we examined the performance of the BAP design under varying maximum sample sizes of 300 and 420 (results omitted for brevity), and found that our previous conclusions based on a maximum sample size of 360 remained consistent across these alternative sample sizes.

8 Concluding Remarks

BAP is an adaptive multi-stage multi-dose platform design and it possesses two important features that are distinctive from existing dose-ranging methods. First, using a shared arm facilitates information sharing across different agents, leading to higher generalized power and smaller sample sizes. Second, BAP incorporates early longitudinal biomarker measurements to predict long-term, temporarily unobservable responses, and thus allows for real-time go/no-go decisions without any trial delay. Our simulation studies demonstrate the advantages of BAP in terms of power for detecting a dose-response effect or clinical relevance, identification of an MED, and savings in sample size.

The BAP design and its variants can be applied in several practical settings. When the primary endpoint is short-term, a simplified version based on the $E_{\max}$ model without longitudinal biomarkers can still improve efficiency and reduce sample size. BAP is also suitable for multi-arm, multi-dose trials, including arms with fixed doses or drug combinations, by modifying the model to remove dose-response terms or include drug-drug interactions. Furthermore, BAP can be extended to collaborative drug development programs involving multiple sponsors, such as those described in Mu et al. (2024), by incorporating site-specific parameters and sharing summary-level information via Bayesian hierarchical models. We also suggest three directions for further research. First, BAP makes parametric assumptions to quantify the associations among treatments, biomarkers, and responses. Although our simulations suggest that BAP has stable operating characteristics under the model-misspecification scenarios considered, it is worth investigating whether there is an additional gain from using more flexible models, such as the MCP-Mod or non-parametric methods. Second, there may be optimal allocation procedures that can further enhance power, trial efficiency, or MED identification accuracy. For example, the D-optimality objective (Padmanabhan et al., 2010) and the concept of the MED-optimal design (Dette et al., 2008) could be applied to BAP. Finally, BAP can be generalized to incorporate both early reads-outs of the long-term response (Van Lancker et al., 2019) as well as multiple biomarkers. Bayesian variable selection priors can be used to identify potentially useful biomarkers for decision making.

Supplementary Materials

The supplementary materials include the computational steps for the proposed model, the configurations of the proposed BAP and competing methods, additional simulation settings and results, and the results of comprehensive sensitivity analyses.

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Data Availability Statement

No datasets were generated or analyzed in the study.

Disclosure Statement

The authors report there are no competing interests to declare.

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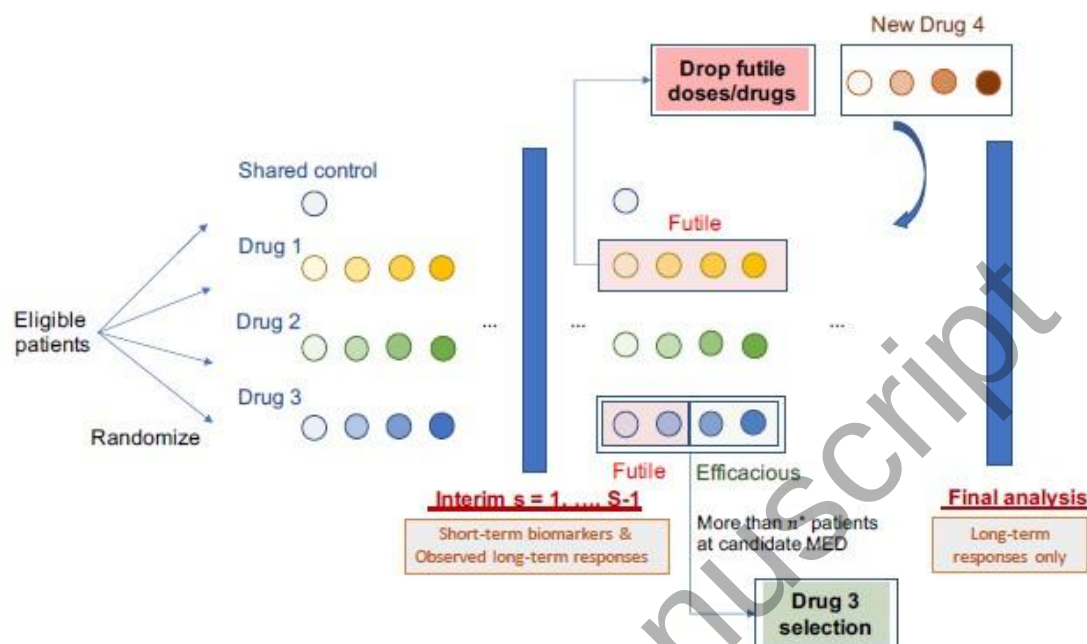
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(a) Trial schema



(b) Structural relationships

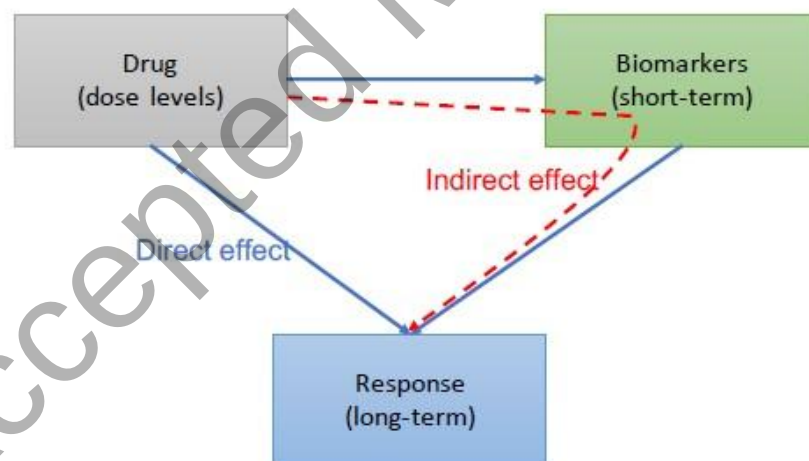


Figure 1: (a): Flowchart for the proposed platform design. n^* is a prespecified sample size cutoff for an early futility or superiority decision for all doses of one agent. Both short-term biomarker and observed long-term response endpoints contribute to the interim decision making. At the end of the trial, only completely-observed, long-term endpoints are used for the final analysis. (b): Structural relationship between the agent (or dose levels), biomarkers, and primary response. The agent has a direct effect on response and an indirect effect on response through the biomarkers.

	Dose-response detection (%)			Overall α	Patient allocation (%)				Total	Duration
Methods	Agent 1	Agent 2	Agent 3	or $1 - \beta$	Control	Agent 1	Agent 2	Agent 3	N	(in months)
Scenario 1										
ANOVA	1.6	1.8	1.7	5	19.9	26.7	26.7	26.7	360	60.0
Iso	1.7	1.7	1.6	5	33.1	22.3	22.3	22.3	360	77.4
Naive	2.2	2	2.1	5.1	20.2	26.5	26.6	26.6	336.7	56.9
Oracle	1.8	1.8	1.8	4.9	20	26.6	26.6	26.8	347.6	76.1
ER	1.8	1.7	1.5	4.5	20	26.6	26.7	26.7	338.9	57.4
Proposed	1.8	1.8	1.9	5	20.1	26.7	26.6	26.6	338.2	57.4
Scenario 2										
ANOVA	1.7	1.7	1.5	4.8	20	26.7	26.7	26.7	360	60.0
Iso	1.7	1.6	1.9	5.1	33.1	22.3	22.3	22.3	360	77.4
Naive	1.8	2.1	1.9	4.8	20.2	26.6	26.6	26.6	335.8	56.8
Oracle	1.7	1.8	2.1	5.1	20	26.6	26.7	26.7	347.5	76.1
ER	1.7	1.6	1.7	4.6	20	26.7	26.7	26.7	339.8	57.5
Proposed	1.7	1.9	2.0	5.1	20	26.6	26.7	26.7	339.2	57.4
Scenario 3										
ANOVA	1.4	1.8	1.7	4.8	20	26.6	26.7	26.6	360	60.0
Iso	1.8	1.6	1.7	5	33.1	22.3	22.3	22.3	360	77.4
Naive	2.1	1.8	2.2	4.9	20.1	26.6	26.6	26.6	336.3	56.9
Oracle	1.7	1.7	1.9	4.9	20	26.6	26.7	26.6	348	76.2
ER	1.8	2	2	5.1	20	26.6	26.7	26.7	341.7	57.8
Proposed	1.8	1.8	1.9	4.9	20	26.6	26.7	26.6	341.9	57.8
Scenario 4										

ANOVA	5.3	5.9	5.5	15.8	20	26.7	26.7	26.6	360	60.0
Iso	30.8	31.4	31.4	67.6	32.6	22.4	22.5	22.5	360	77.3
Naive	21.2	22.4	21.9	37.6	20	26.6	26.6	26.7	354.3	59.7
Oracle	35	35.6	35.4	55.2	20	26.6	26.6	26.7	354.1	77.1
ER	34.7	34.6	35	54.5	20	26.7	26.7	26.6	352.6	59.5
Proposed	34.4	34.2	34.5	54.2	20	26.7	26.7	26.7	352.1	59.4
Scenario 5										
ANOVA	3.5	3.5	3.6	10.1	20	26.7	26.7	26.6	360	60.0
Iso	30.1	30.2	30.4	66.1	32.6	22.5	22.5	22.4	360	77.4
Naive	21	20.6	20.5	50.2	20	26.7	26.7	26.7	354.2	59.7
Oracle	35.6	35.1	35.5	55.4	20	26.6	26.7	26.7	354.2	77.1
ER	34.6	35.3	35	54.6	20	26.7	26.7	26.7	353.6	59.6
Proposed	33.6	34.4	34.5	54.2	20	26.7	26.7	26.6	353.1	59.6
Scenario 6										
ANOVA	1.6	86.8	96.7	99.6	20	26.7	26.7	26.7	360	60.0
Iso	1.6	86.1	99.3	99.9	33	22.3	22.7	22.1	360	77.4
Naive	2.2	95.4	100	100	20.1	25.7	27.1	27.1	348.2	58.8
Oracle	2.4	98.9	100	100	20	26.3	26.9	26.7	345.4	75.8
ER	2.1	97.2	100	100	20	25.4	27.3	27.3	346.7	58.6
Proposed	2.2	98.3	99.9	100	20	25.8	27.3	27	342.5	58

	Dose-response detection (%)			Overall α	Patient allocation (%)				Total	Duration
Methods	Agent 1	Agent 2	Agent 3	or $1 - \beta$	Control	Agent 1	Agent 2	Agent 3	N	(in months)
Scenario 7										
ANOVA	100	99.4	9.8	100	20	26.7	26.7	26.7	360	60.0
Iso	100	99	58.4	100	34.2	20.9	22.5	22.5	360	77.4
Naive	100	99.8	46.9	100	20	24.6	27.7	27.7	341.3	57.8
Oracle	100	100	55.1	100	20	22.4	27.9	29.7	318.9	71.7
ER	100	100	53.3	100	20	26.7	26.7	26.6	354.8	59.8
Proposed	100	100	63.4	100	20	23	28.2	28.8	326.5	55.6
Scenario 8										
ANOVA	100	25.5	100	100	20	26.7	26.6	26.7	360	60.0
Iso	100	0	100	100	34.2	21.6	21.8	22.4	360	77.4
Naive	100	0	100	100	20.4	29.6	20	30	314.4	53.7
Oracle	100	0	100	100	20	26.3	21.2	32.5	283.9	66.6
ER	100	0	100	100	20	30	19.8	30.1	314.2	53.7
Proposed	100	0	99.8	100	20	27.9	20.2	31.9	290.5	50.0
Scenario 9										
ANOVA	60.7	100	61.5	100	20	26.6	26.7	26.7	360	60.0
Iso	59.6	99.9	52.7	100	32.8	22.7	22	22.5	360	77.4
Naive	78	100	57.8	100	20.1	26.7	26.5	26.7	353.4	59.6
Oracle	89.3	100	77.4	100	20	27.3	25.5	27.2	340.8	75.0
ER	82.4	100	59.6	100	20	26.7	26.7	26.6	354	59.7
Proposed	87.5	100	75.9	100	20	27.2	25.9	26.9	343.6	58.2
Scenario 10										

ANOVA	69.4	99.9	33.1	100	20	26.7	26.7	26.7	360	60.0
Iso	79.1	100	98	100	35.1	22.7	21	21.2	360	77.4
Naive	94.4	100	99.9	100	20	27	26	27	350.4	59.1
Oracle	97	100	99.9	100	20.1	27.5	24.7	27.8	340.5	75.1
ER	94.7	100	99.8	100	20	26.7	26.7	26.7	354.8	59.8
Proposed	96.7	100	99.8	100	20	27.3	25.2	27.5	343.2	58.1
Scenario 11										
ANOVA	100	90	100	100	20	26.7	26.7	26.6	360	60.04
Iso	100	90.6	100	100	34.4	21.7	22.6	21.3	360	77.4
Naive	100	98.4	100	100	20	26.6	26.9	26.5	352.1	59.4
Oracle	100	99.6	100	100	20	26.3	29.6	24.2	314.9	71.3
ER	100	99.5	100	100	20	26.7	26.7	26.7	354.9	59.9
Proposed	100	99.7	100	100	20	26.6	28.9	24.6	324.2	55.2
Scenario 12										
ANOVA	64	20.5	12.7	74.7	20	26.7	26.7	26.7	360	60.0
Iso	82.7	0.5	14	85	33	22.6	22	22.4	360	77.4
Naive	93.2	0	13.6	93.2	20	27.9	24.2	27.9	337.6	57.2
Oracle	93.2	0	13.6	93.2	20	27.9	24.2	27.9	337.6	74.6
ER	96.6	0.1	22	96.6	20	28.1	24	27.8	336.1	57
Proposed	97.7	0.1	26.6	97.7	20	28.1	24	27.9	334.2	56.7